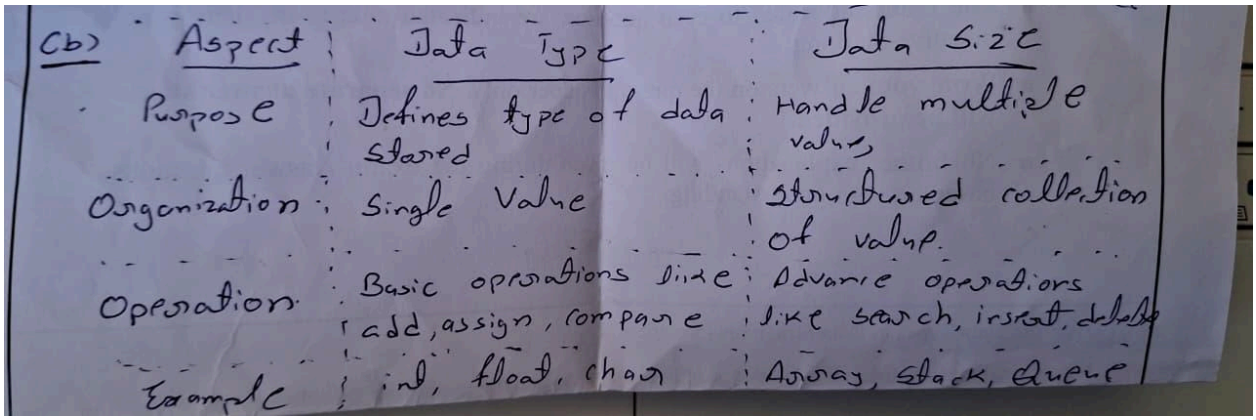
IT-205 First In-Sem Exam Paper Checking Style:

Q-1)

a) Primitive datastypes desigfned to store a single, independent value. So problem arise,
- Organization: To store each entry we need to make new variable. Let's say to save 1000 reading we need to make 1000 variable.
- Relation: Can't map the reading to time of arrival or came from which sensor and order of arrival.
(explain 2 limitations precisely for a full mark. Don't randomly state increas in time and space complexity. State why?)

b)



- For full marks, include all 3 aspect.

c) Array / Dynamic Array / Vector.
- State O(1) for accessing the element by index and for updating (Justification)
(For all 3 requirements, there is only one answer - ARRAY (1 Mark), and justification is O(1) (1 Mark))

Q-2)

a) 0.5 Marks if Explains Concepts of Data Abstraction and Functional Abstraction and 0.5 marks if Shows How They work Together in Software design
b) Provides Constructors , Accessor Methods, Modifier Methods behaviour purpose than 2 Marks .
c) For Stating Advantages 1 Mark and For the Reason of why Internal Representation of CourseList should be hidden from the user 1 Mark

Q-3)

a) (address value should be printed not data value)

a. if 7800 is mentioned then **2 marks** with explanation

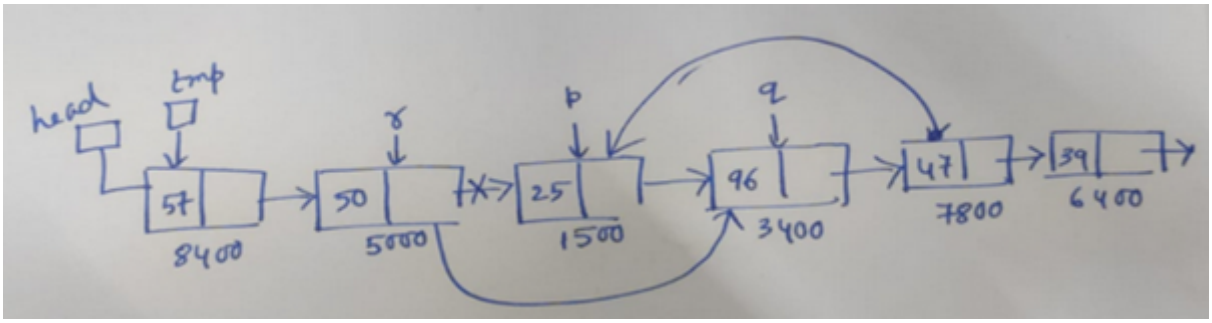
b. if 96 is mentioned then **0.5 marks**

b) (address value should be printed not data value)

a. if 1500 is mentioned then **2 marks** with explanation

b. if 47 is mentioned then **0.5 marks**

c)



1 mark if correct Linked List drawn
0 mark if incorrect Linked List drawn

Q4)

a) general formula - 1 mark

A(10) if correct - 1mark

b)Marks given only when total is correct
3n-2

c)Marks given only if correct eqn written

(c) Upto row i-1-th row —- 1+2+3+... +i = (i(i + 1)/2)

i-th row —- j − (n − 1 − i)

Address(M[i][j]) = B + [(i(i + 1)/2) + (j − (n − 1 − i))] × w

Marks deducted if 'w' not multiplied.

d)correct -O(n)